

Saying “I’m Not Sure” to Young Learners: A Protocol for Designing and Expert-Reviewing Uncertainty Cues in a Children’s AI Tutor

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Study protocol. No results yet. Not peer reviewed.

Abstract

If an AI tutor should sometimes decline to answer, it also needs a way to say so that a child can understand. Research with adults shows that first-person expressions of uncertainty can reduce overreliance on a language model, and research with children shows that they form views about conversational agents’ intelligence and trust, but how a tutor should express uncertainty to young learners remains largely undesigned. This protocol describes a design study for learners of about ages 8 to 11. I will build a scripted, clickable prototype of one short science lesson in five variants that differ only in the uncertainty cue: no cue, a plain-language hedge, a visual confidence meter, a character who shows it is thinking, and an explicit “I don’t know” with a hand-off to a teacher. The variants will be evaluated without children, through a heuristic review and walkthroughs by three to five adult experts. The output is a set of candidate cues and design risks, ready for a later study with children under ethics review.

Keywords: uncertainty communication; children; AI tutors; abstention; trust calibration; heuristic evaluation; prototyping

1. Introduction

My earlier work found that a retrieval-augmented question-answering system never declined to answer, even on questions it had no basis for (Choi, 2026). If a tutor *should* sometimes say it does not know, a second question follows: how should it say it, especially to a child?

Children are a demanding audience for this. A cue must be understandable at their reading level, it should invite a useful next step such as checking a book or asking a teacher, and it should not make the tutor seem broken or untrustworthy in general. This study designs candidate cues and gathers expert critique before any child is involved.

How can an AI tutor tell a young learner that it is unsure in a way that prompts checking rather than blind acceptance or blanket distrust?

2. Background

Druga et al. (2017) studied how 26 children aged 3 to 10 interacted with four agents, including Amazon Alexa and Google Home, and asked them about trust, intelligence, and personality; they identified four themes, including perceived intelligence and understanding. Ruan et al. (2020) found, with 72 children in grades 3 to 5, that embedding a math task in a narrative boosted engagement and that adding a tutoring chatbot improved learning outcomes. With adults, Kim et al. (2024) found in a pre-registered experiment (N = 404) that first-person uncertainty expressions such as “I’m not sure, but...” decreased agreement with a fictional LLM-infused search engine and increased participants’ accuracy. Heuristic evaluation (Nielsen & Molich, 1990) offers a way to find interface problems without end users, which suits an early design stage.

Table 1. Prior work and what this study adds.

Work	Finding	What it leaves open
Druga et al. (2017)	Children form views of agents' intelligence and trust	How agents should signal their limits
Ruan et al. (2020)	A narrative tutor with chatbot help supported children's learning	The tutor's own uncertainty as a design variable
Kim et al. (2024)	First-person uncertainty reduced adults' overreliance	Whether similar cues work for children
Nielsen & Molich (1990)	Heuristic evaluation finds problems without end users	Heuristics specific to children and AI uncertainty

3. Research questions

1. **RQ1.** What forms can an AI tutor's uncertainty take for young learners, in words, visuals, character behavior, or a hand-off to a person?
2. **RQ2.** Which forms do experts judge most likely to be understood by children and to prompt checking rather than blind acceptance?
3. **RQ3.** What risks do experts see, such as children losing trust entirely or concluding that the tutor is broken?

4. Method

4.1 Design space

Each variant answers the same questions in the same lesson, so the cue is the only thing that changes (Table 2).

Table 2. Uncertainty cue variants.

Variant	Cue	What the child sees or hears
A. No cue (control)	None	A confident answer
B. Plain words	First-person hedge	"I'm not sure about this one, but I think..."
C. Visual meter	Confidence picture	A small gauge icon next to the answer
D. Character	The guide shows it is thinking	The guide character pauses and suggests checking the book
E. Hand-off	Abstain and redirect	"I don't know. Let's ask your teacher or look it up together."

4.2 Prototype

The prototype is scripted rather than driven by a live model, so every variant shows identical answers. It covers one short lesson (for example, why leaves change color) with six questions: three the tutor answers correctly, two it answers incorrectly, and one it cannot answer. It will be built as a clickable Figma prototype or a single web page with a variant switcher. The guide character will be an original drawing made for this study.

4.3 Evaluation

1. **Heuristic review.** I will score each variant against Nielsen's heuristics plus three written for this problem: the cue is understandable at the target reading level; the cue invites a next action, such as checking; and the cue does not make the tutor seem broken.

2. **Expert walkthroughs.** Three to five adult experts (teachers or HCI researchers) will each click through all variants in a randomized order for about 30 minutes and answer the same questions for each: would a child understand this, what would the child do next, and what could go wrong.

Open item. Expert walkthroughs may count as human subjects research. Before inviting experts, I will ask my university's IRB office whether a determination is needed. If the answer comes too late, I will run the heuristic review alone and report it as a single-author review.

5. Analysis plan

- **RQ1:** the design space itself, documented with screenshots of each variant.
- **RQ2:** heuristic scores per variant, and expert judgments summarized by variant with counts.
- **RQ3:** expert comments coded by theme; every reported theme names how many experts raised it.

6. Ethics and limitations

No children take part in this study. Any later study with children will require ethics review and parental consent.

- Adult judgments about children are predictions, not evidence of how children respond.
- A scripted prototype cannot show how cues behave with a real model's errors.
- Three to five experts give direction, not generalizable findings.

7. Declarations

Origin of this work. This study extends the abstention problem described in Choi (2026) to the design of uncertainty cues for children.

Use of AI tools. I used Claude (Anthropic), an AI assistant, to verify references against Crossref records and to draft text and tables.

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Competing interests. None.

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